

Omniverse Compute (OMC) Decentralized GPU Computing Network Whitepaper (v2.0)

Version: v2.0 | Network Support: BNB Chain Ecosystem & Layer 2 Rollup Dedicated Scaling
Network | Native Token: OMC (BEP-20)

Executive Summary

Omniverse Compute (OMC) utilizes the Decentralized Physical Infrastructure Network (DePIN) paradigm to build a high-throughput, low-cost, transparent, and censorship-resistant global GPU compute sharing grid. Driven by explosive demand in artificial intelligence (AI) large language model (LLM) training, multimodal inference, real-time 3D cloud rendering, and embodied AI, traditional centralized cloud computing infrastructure faces severe challenges including prohibitive CapEx, strict quota restrictions, and cloud vendor lock-in. OMC leverages BNB Chain as its root settlement layer alongside a dedicated Layer 2 Rollup for high-frequency task scheduling and per-second/per-task micropayments, delivering a seamless, global plug-and-play compute infrastructure.

Section 1: Background, Industry Outlook & Commercial Value

1.1 The Inevitable Boom & Economics of DePIN Compute Networks

- **Compute as the New Oil:** As AI transitions from discriminative to generative models, computing demand for LLMs, Sora-grade video generation, and spatial simulation is exploding exponentially. Traditional data center expansion faces severe power grid and hardware supply chain bottlenecks.
- **Reshaping Compute Economics:** Globally, hundreds of thousands of consumer-grade GPUs (e.g., RTX 4090/3090), data center idle capacity, and post-PoW mining rigs remain underutilized. DePIN reduces marginal cost to near zero, providing 50%–70% direct cost savings for compute requesters.
- **Trillion-Dollar Ecosystem Flywheel:** Connecting Compute Providers, AI Developers, DeAI Autonomous Agents, and VFX Studios into a unified tokenized ecosystem with perpetual deflationary value capture.

1.2 Target Market Demand & Four Client Personas

- **AI Startups & Independent Developers:** Lacking big-tech enterprise discounts, facing high hourly API costs and GPU quota limits. Use cases include LLM fine-tuning, LoRA training, and model inference. Acquisition via HuggingFace, GitHub, and AI hackathons.
- **Web3 AI & DeAI Ecosystem Projects:** Requiring censorship-resistant, natively crypto-paid compute networks for 24/7 Agent inference and ZK-ML proof generation. Acquisition via BNB Chain and EVM DeAI alliances.
- **Cloud Rendering, 3D CG & Metaverse Studios:** Facing extreme compute spikes for VFX, Unreal Engine 5 real-time rendering, Blender offline batch rendering, and Gaussian Splatting 3D reconstruction.
- **Hardware Providers & Mining Transition Operations:** Connecting idle hardware (RTX 4090s, mining rigs, data center servers) into OMC protocol to generate automated token yield.

Section 2: Technical Architecture & System Design

2.1 Layered Architecture

- L1 Root Settlement Layer: BNB Chain handles OMC token issuance, staking smart contracts, DAO governance, and final dispute arbitration.
- L2 Rollup Orchestration Layer: Dedicated Layer 2 Rollup executing high-frequency node heartbeats, task allocation state mapping, state channel settlements, and low-cost micropayments.
- Off-Chain Compute Grid: Containerized Docker/Kubernetes sandbox supporting CUDA/ROCM hardware acceleration, NVLink inter-GPU communication, and P2P data transport.

2.2 Task Matching Engine & Scheduling Algorithm

The decentralized scheduling engine assigns tasks based on Node VRAM, Floating Point Performance (TFLOPS), Network Bandwidth, Geographic Latency, and Historical Reputation Score using a weighted bipartite matching algorithm.

$\text{Score} = 0.35 \times \text{TFLOPS} + 0.25 \times \text{VRAM} + 0.15 \times \text{Bandwidth} + 0.25 \times \text{Reputation}$

Python

```
# OMC Decentralized Compute Scheduling Node Matching Logic
def calculate_node_score(node):
    weight_flops = 0.35
    weight_vram = 0.25
    weight_bandwidth = 0.15
```

```
weight_reputation = 0.25
```

```
score = (node.tflops * weight_flops) + \  
        (node.vram_gb * weight_vram) + \  
        (node.bandwidth_mbps * weight_bandwidth) + \  
        (node.reputation_score * weight_reputation)  
return score
```

Section 3: Node Verification, Staking & Slashing Mechanisms

3.1 Dual-Verification System

- Redundant Computation Verification: High-precision tasks are dispatched to 2–3 independent nodes simultaneously to compare output feature hashes.
- zk-SNARKs Sampling Verification: Lightweight zero-knowledge proofs enable on-chain verification of execution traces without compromising data privacy.

3.2 Staking & Slashing Rules

- Staking Threshold: Nodes must stake OMC proportional to hardware tier before receiving jobs.

$\text{Staking_Min} = \text{Base_Deposit} \times \text{Card_Tier_Multiplier}$

- Slashing Triggers: Fraudulent outputs, task timeouts, unexpected offline dropouts, or fake heartbeat signals.
- Slashing Re-allocation: 50% compensates affected task requesters, 30% goes to DAO Treasury, and 20% is permanently burned:

$\text{Burn_Amount} = \text{Slashed_Tokens} \times 20\%$

Section 4: Tokenomics & Deflationary Value Capture

4.1 Token Distribution (1,000,000,000 OMC Fixed Supply)

- Token Name / Symbol: Omniverse Compute (OMC)
- **Token Standard:** BEP-20 (BNB Chain Ecosystem)
- Max Total Supply: 1,000,000,000 OMC

Category	Allocation (%)	Amount (OMC)	Vesting Schedule
Compute Mining & Node Rewards	50%	500,000,000	0% at TGE, linear decay release over 10 years based on compute contributions (halving every 2 years).
Investors (Institutional/Private/Airdrop)	20%	200,000,000	Institutional 10% (TGE 5%, 6m cliff, 24m monthly); Private 8% (TGE 10%, 3m cliff, 18m monthly); Airdrop 2% (TGE 100%).
Ecosystem & Grants	15%	150,000,000	5% at TGE, 6-month lockup, 36-month monthly linear vesting for grants & API subsidies.
Team & Core Contributors	10%	100,000,000	0% at TGE, 12-month cliff, 24-month monthly linear vesting.
Treasury & Liquidity	5%	50,000,000	50% unlocked at TGE for DEX/CEX initial liquidity; remaining managed by DAO Treasury.

4.2 Deflationary Mechanics

1. Protocol Fee Burn: A 3% protocol service fee is levied per job settlement; 30% of this fee is automatically bought back and burned via smart contracts, while 70% enters the Treasury to reward high-reputation nodes.
2. Dual Burn Engine: Service fee burn + Slashing penalty burn constantly decrease circulating supply to reinforce OMC value growth.

Section 5: Roadmap & Governance

- **Phase 1 (Q1-Q2 2026):** L2 smart contract deployment, open-source node client, testnet launch, and 2% community airdrop.
- **Phase 2 (Q3-Q4 2026):** Mainnet launch, compute mining activation, HuggingFace & Blender plugin integration.
- **Phase 3 (2027+):** On-chain ZK-ML framework launch, full DAO governance transition, and cross-chain compute settlement expansion.